Graph: (add in the code)

1. Training vs validation accuracy
2. **Training vs validation loss**
3. **Training vs validation precision**
4. **Training vs validation recall**
5. **Training vs validation f1 score**
6. **AUC-ROC curve**
7. **Confusion matrix**

Add in code: model.summary() →> number of parameters

Epoch= 30
Learning rate:0.0001
No of Dropout Layer: 1
Dropout value:0.3
optimizer=Adam
loss='binary_crossentropy'

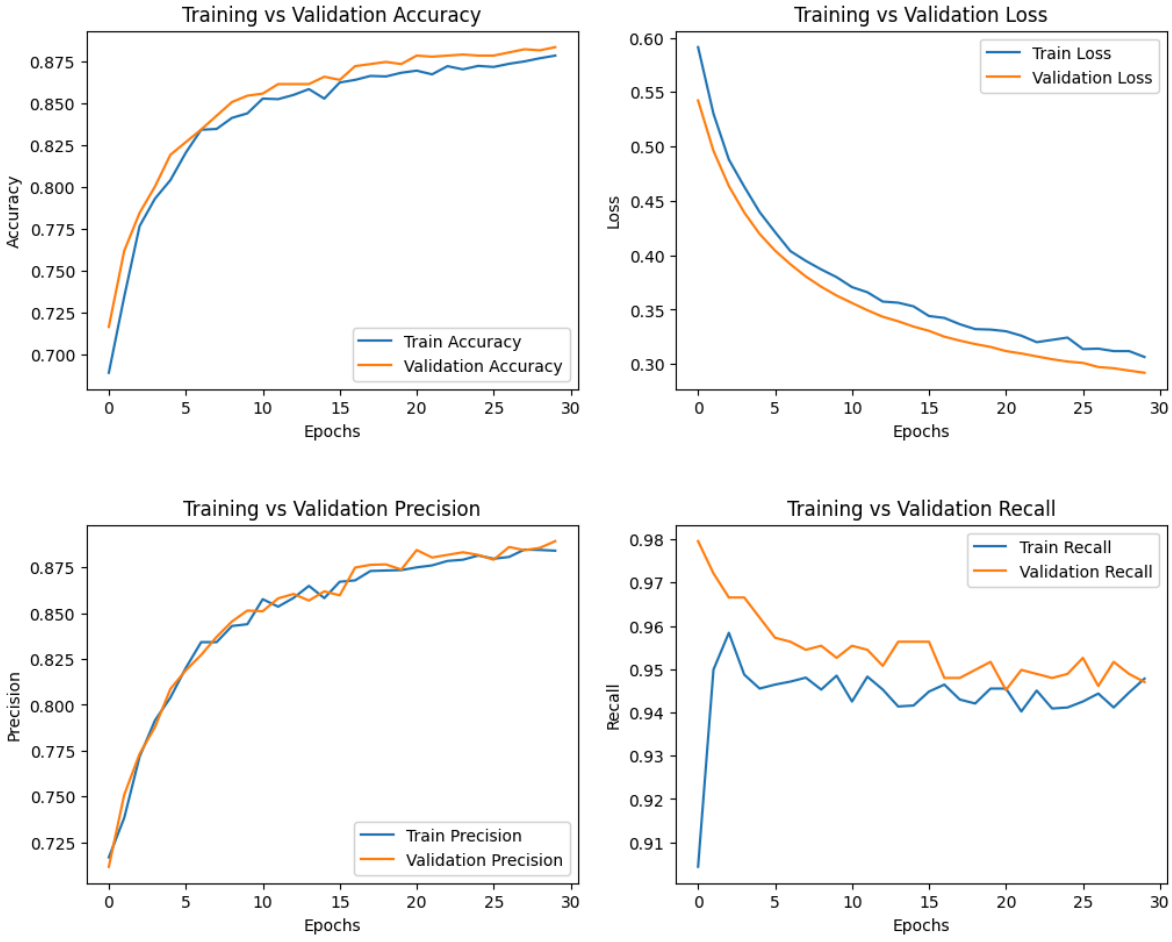
Model	Acc	Pre	Recall	F1	Loss	Running time	Model Parameter (total and trainable)
EfficientNetB0	0.8835	0.8825	0.9563	0.9179	0.2891	4870.15 seconds	total=4,050,852 trainable=1,281 non-trainable=4,049,571

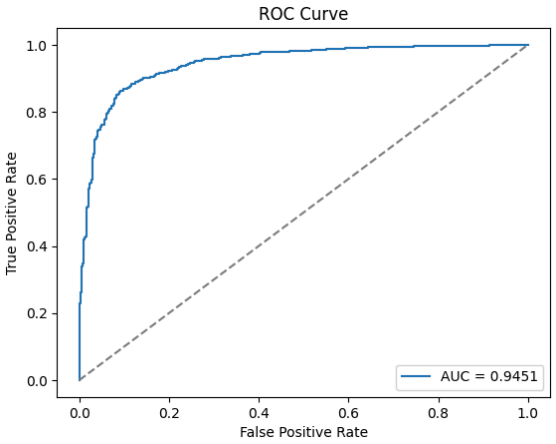
EfficientNetB2	0.8770	0.8747	0.9562	0.9136	0.3131	1552.87 seconds	**Total=** `7,769,978`
							`Trainable: 1,409`
							`Non-trainable: 7,768,569`

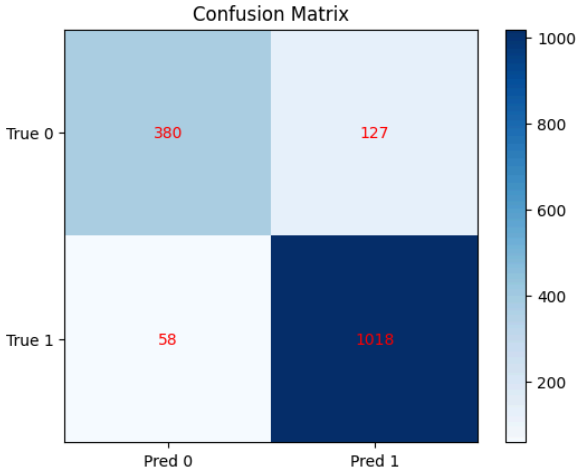
MobileNetV2	0.8681	0.8886	0.9217	0.9048	0.3436	1478.38 seconds	Total: 2,259,265 Trainable: 1,281 Non-trainable: 2,257,984
MobileNetV3Small	0.8082	0.8124	0.9340	0.8690	0.4306	3205.69 seconds	Total: 939,697 Trainable: 577 Non-trainable: 939,120
InceptionV3	0.8348	0.8504	0.9191	0.8834	0.3611	5113.87 seconds	Total: 21,804,833 Trainable: 2,049 Non-trainable: 21,802,784
Vgg16	0.8290	0.8290	0.8988	0.8784	0.3855	4619.67 seconds	Total params: 14,715,201

							Trainable params: 513 Non-trainable params: 14,714,688
Vgg19	0.8270	0.8531	0.9037	0.8777	0.4017	2212.52 seconds	Total params: 20,024,897 Trainable Parameters: 513 Non-trainable Parameters: 20,024,384

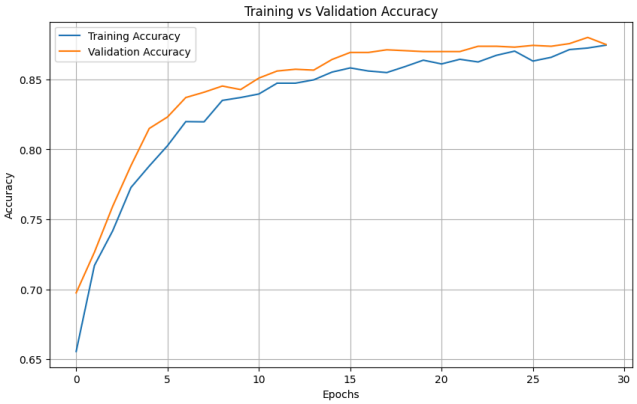
EfficientNetB0:



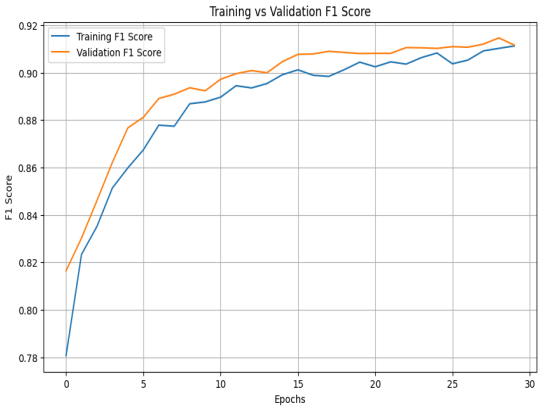


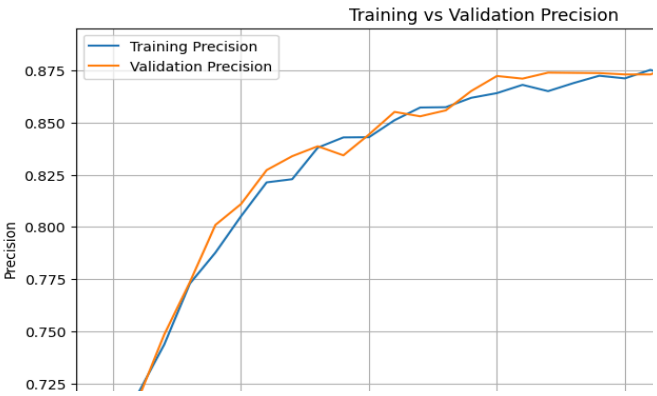


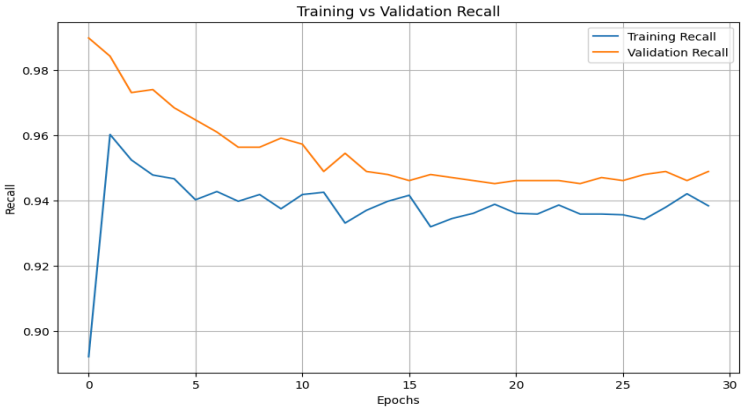
EfficientNetB2:

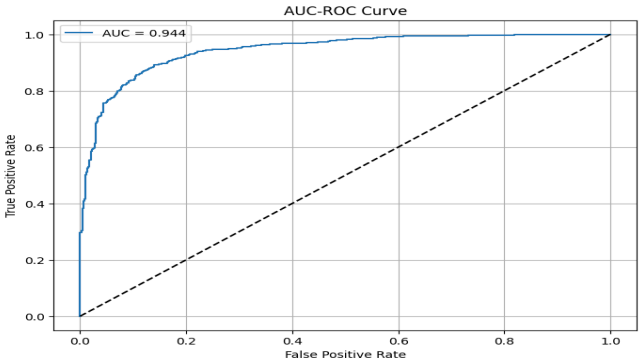


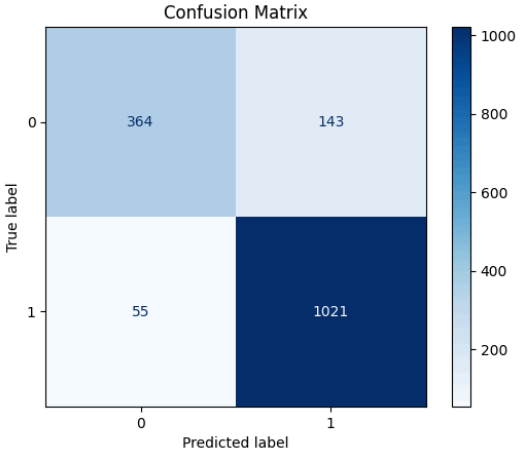








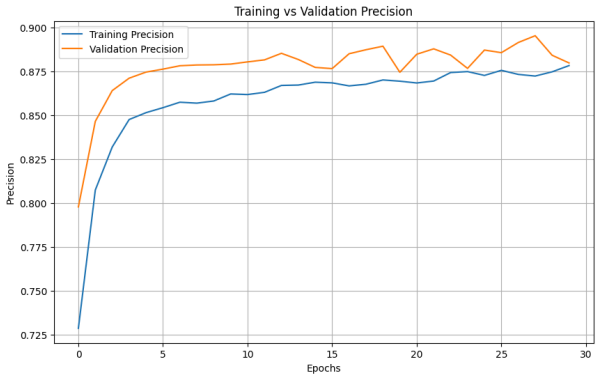


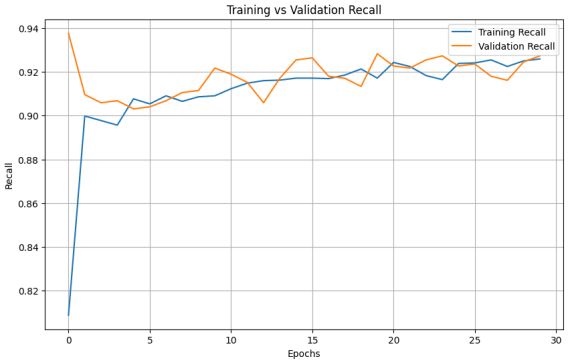


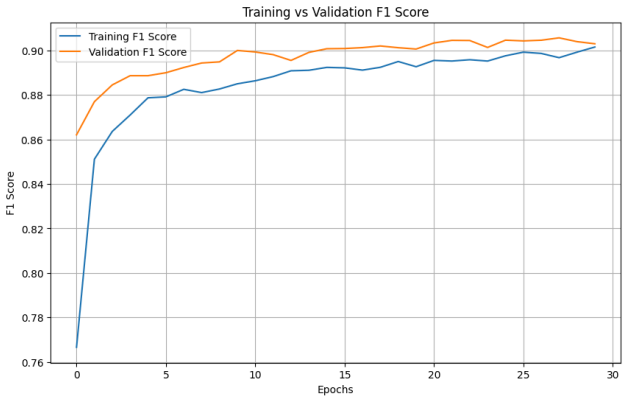
MobileNetV2:

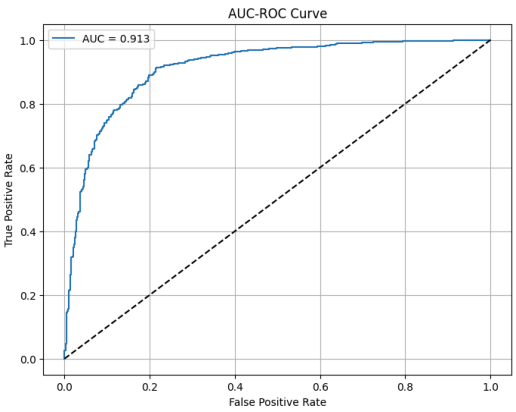

Training vs Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epochs

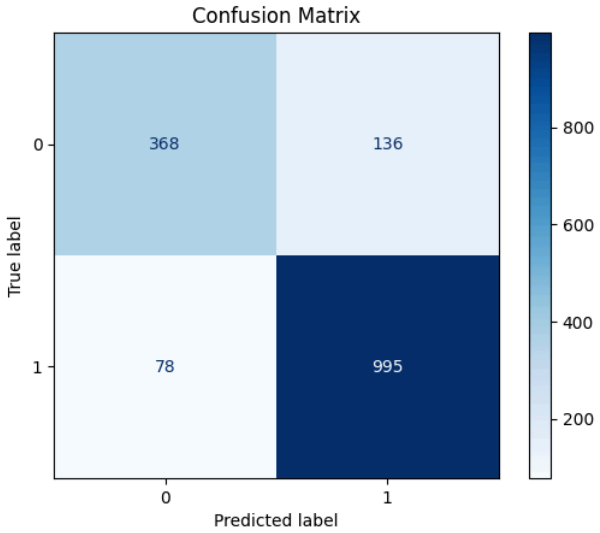

Training vs Validation Loss
Training Loss
Validation Loss
Loss
Epochs


Training vs Validation Precision
Training Precision
Validation Precision
Precision
Epochs

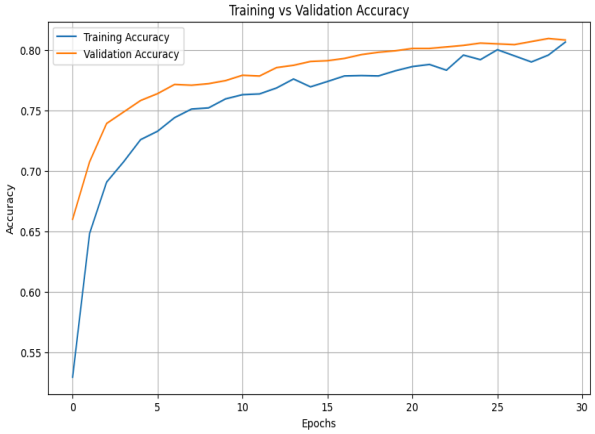

Training vs Validation Recall
Training Recall
Validation Recall
Recall
Epochs

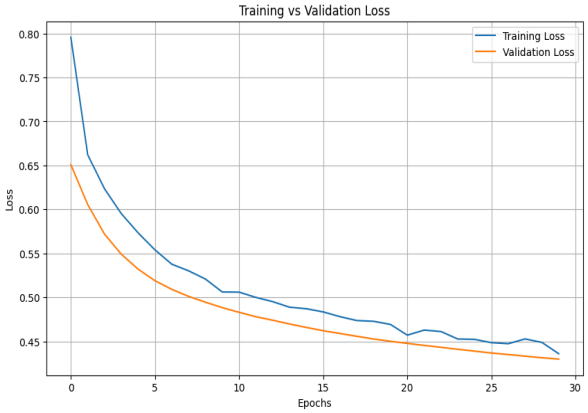

Training vs Validation F1 Score
Training F1 Score
Validation F1 Score
F1 Score
Epochs

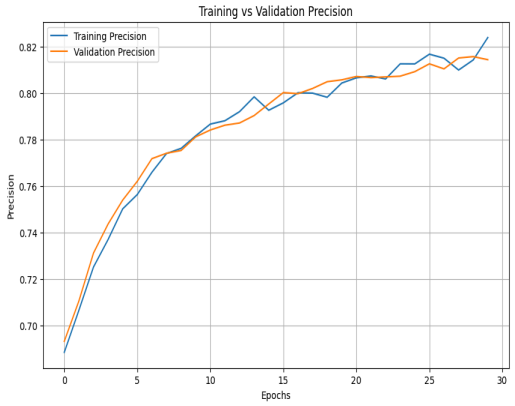

AUC-ROC Curve
AUC = 0.913
True Positive Rate
False Positive Rate

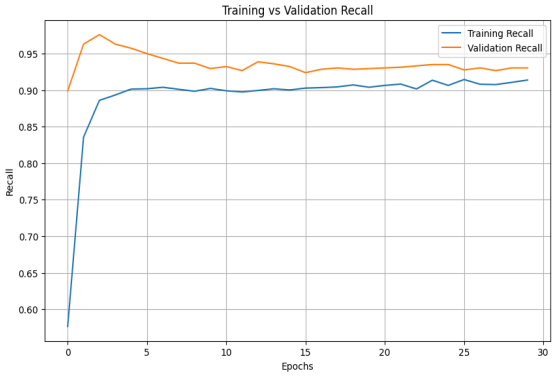


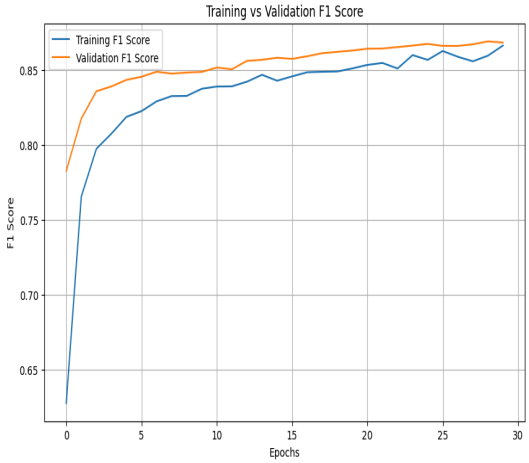
MobileNetV3Small:

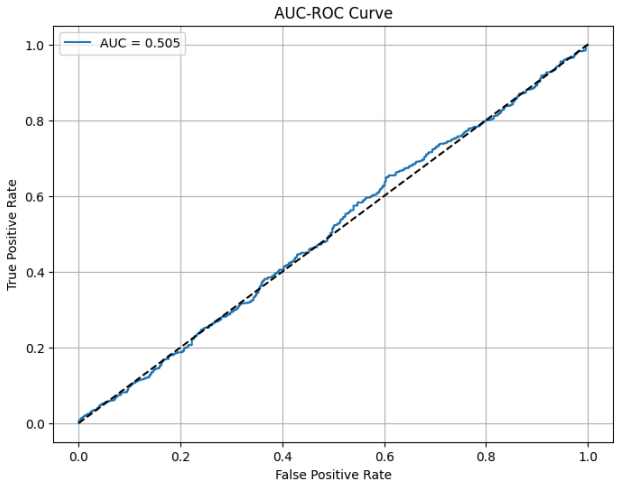


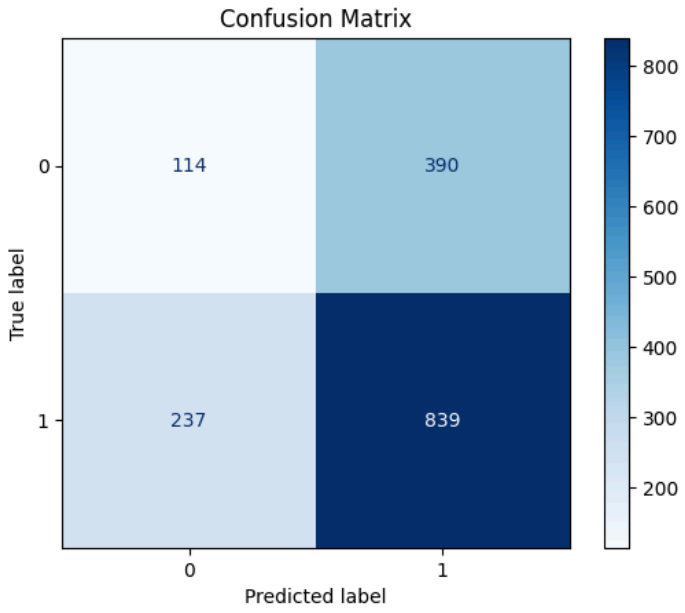



Training vs Validation Precision
Training Precision
Validation Precision
Precision
Epochs

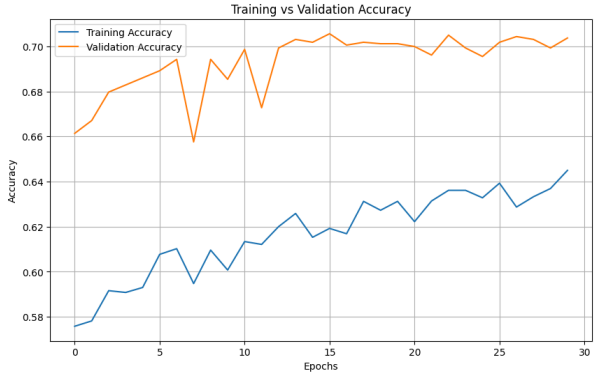

Training vs Validation Recall
Training Recall
Validation Recall
Recall
Epochs

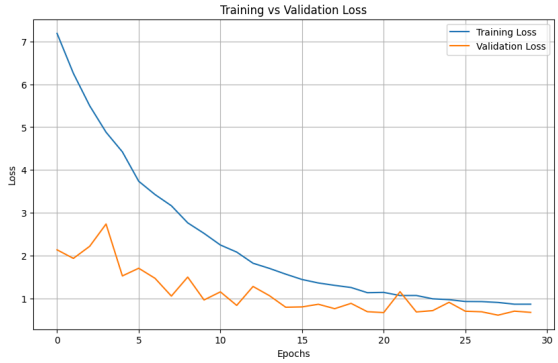

Training vs Validation F1 Score
Training F1 Score
Validation F1 Score
F1 Score
Epochs

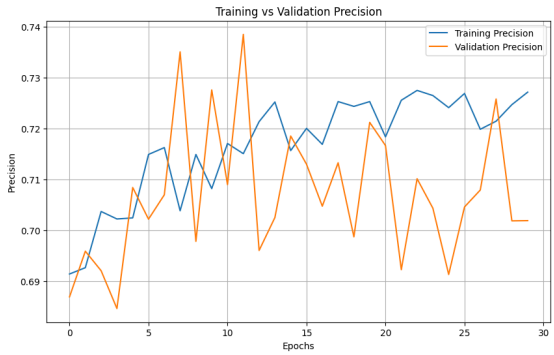

AUC-ROC Curve
AUC = 0.505
True Positive Rate
False Positive Rate

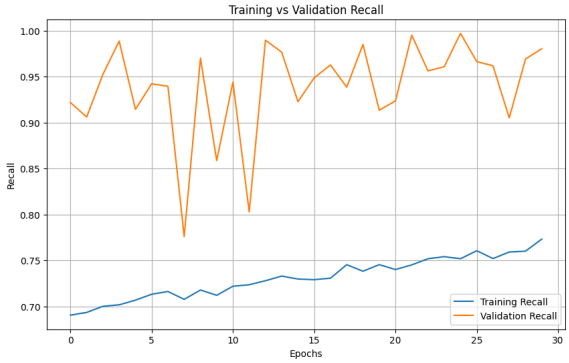

Confusion Matrix
114
390
237
839
True label
Predicted label

InceptionV3:

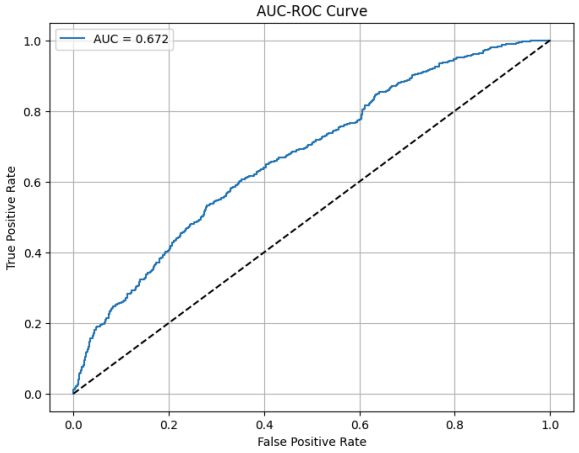


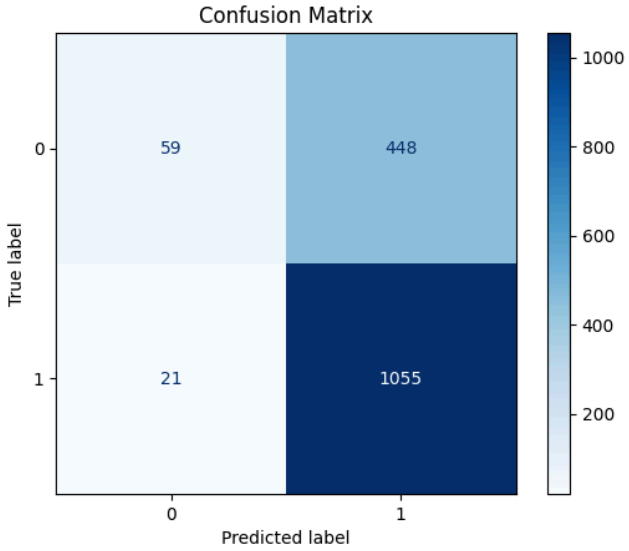




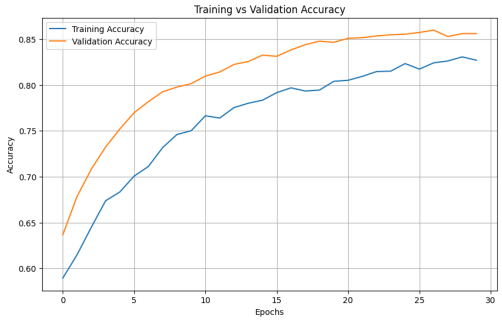


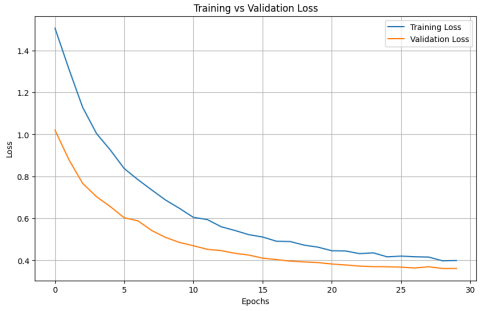

Training vs Validation F1 Score
F1 Score
Epochs
Training F1 Score
Validation F1 Score

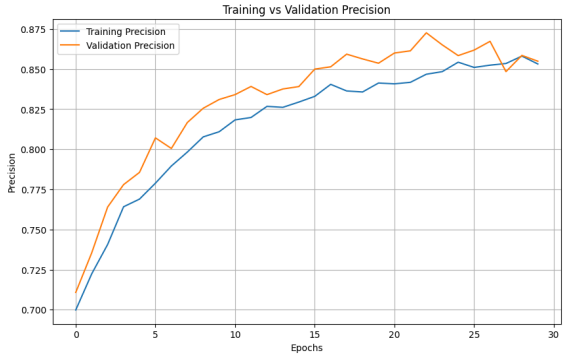

AUC-ROC Curve
AUC = 0.672
True Positive Rate
False Positive Rate

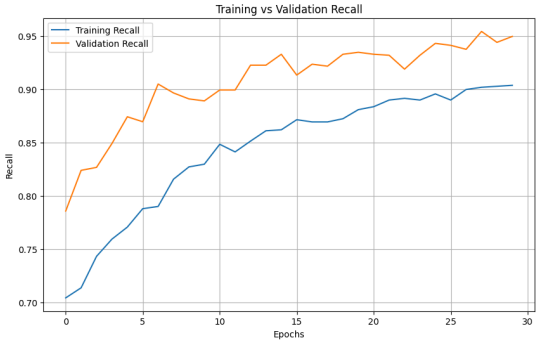

Confusion Matrix
59
448
21
1055
True label
Predicted label

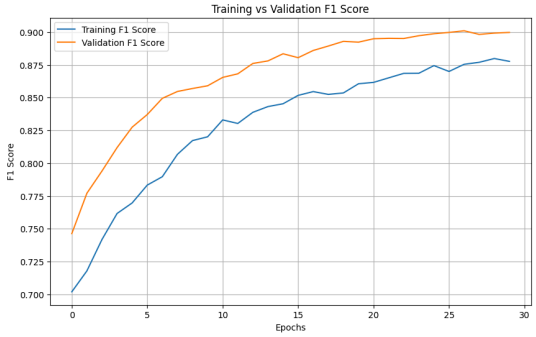
Vgg19:

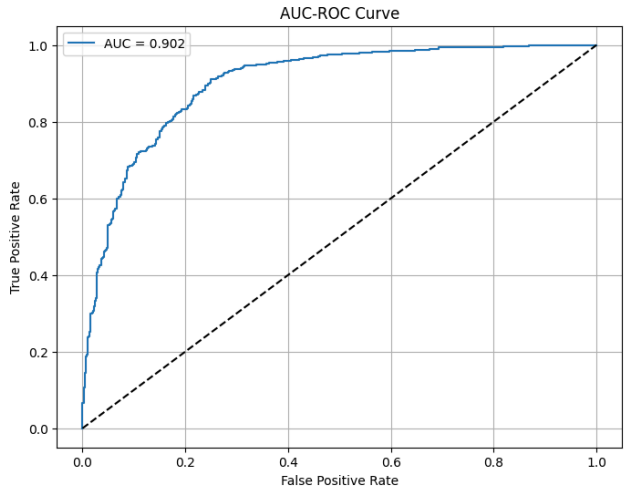

Training vs Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epochs

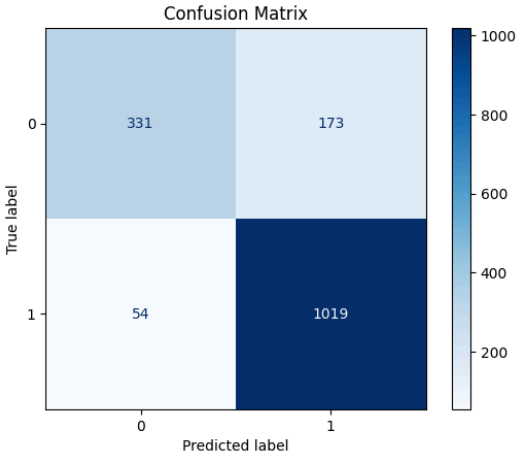

Training vs Validation Loss
Training Loss
Validation Loss
Loss
Epochs


Training vs Validation Precision
Training Precision
Validation Precision
Precision
Epochs

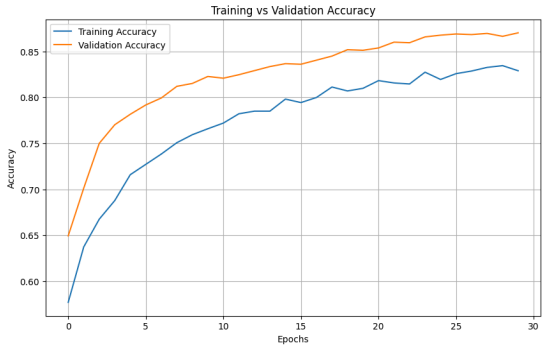

Training vs Validation Recall
Training Recall
Validation Recall
Recall
Epochs

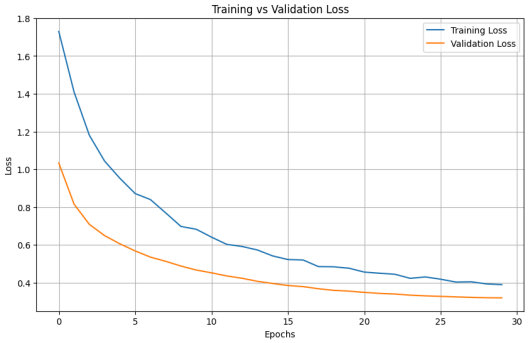

Training vs Validation F1 Score
Training F1 Score
Validation F1 Score
F1 Score
Epochs

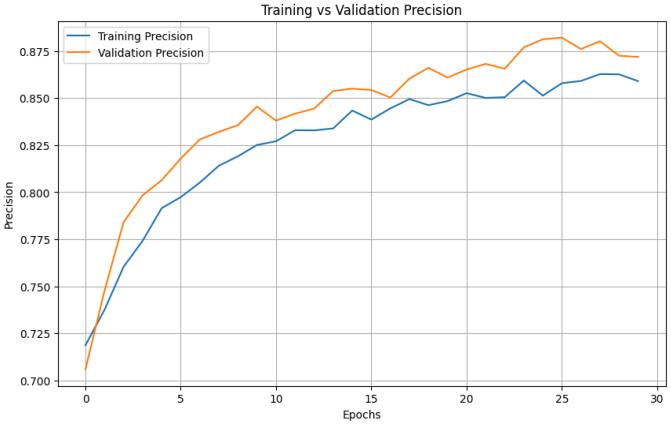

AUC-ROC Curve
AUC = 0.902
True Positive Rate
False Positive Rate

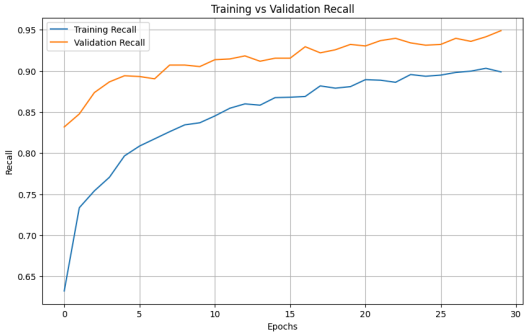

Confusion Matrix
331
173
54
1019
True label
Predicted label

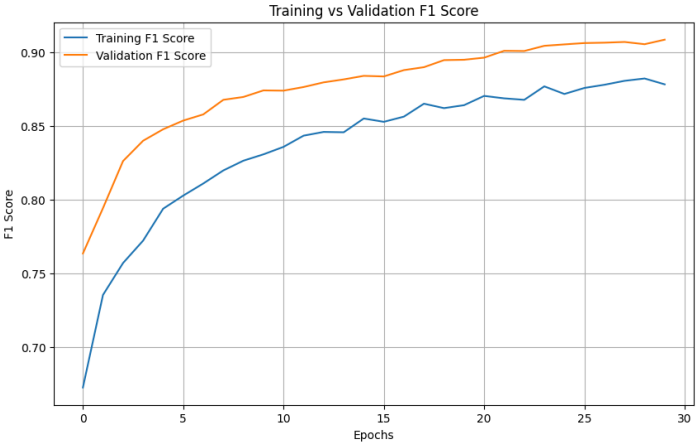
Vgg16:

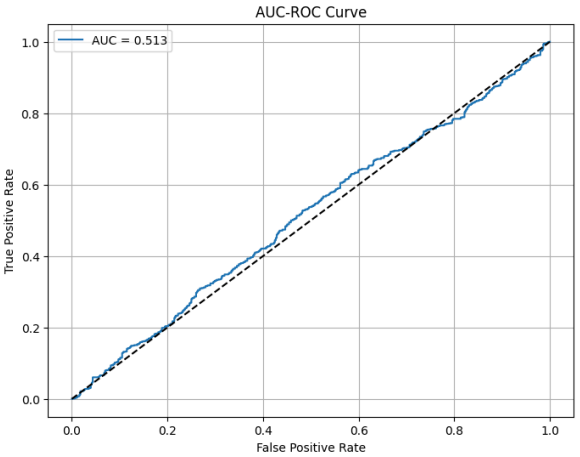

Training vs Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epochs

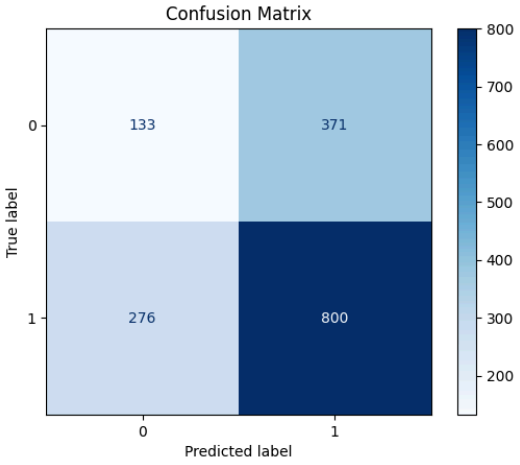

Training vs Validation Loss
Training Loss
Validation Loss
Loss
Epochs


Training vs Validation Precision
Training Precision
Validation Precision
Precision
Epochs


Training vs Validation Recall
Training Recall
Validation Recall
Recall
Epochs


Training vs Validation F1 Score
Training F1 Score
Validation F1 Score
F1 Score
Epochs


AUC-ROC Curve
AUC = 0.513
True Positive Rate
False Positive Rate


Confusion Matrix
133
371
276
800
True label
Predicted label

FINE TUNED PARAMETERS

—----------

No. of UNFREEZ LAYERS: 68(20%)

Frozen Layers: 272 (80%)

Epoch= 30 total Layers: 340 **Learning rate:0.0001** **No of Dropout Layer: 1** **Dropout value:0.3** **optimizer=Adam** **loss='binary_crossentropy'**	Frozen Layers: 272 Unfrozen Layers: 68 Total Layers: 340

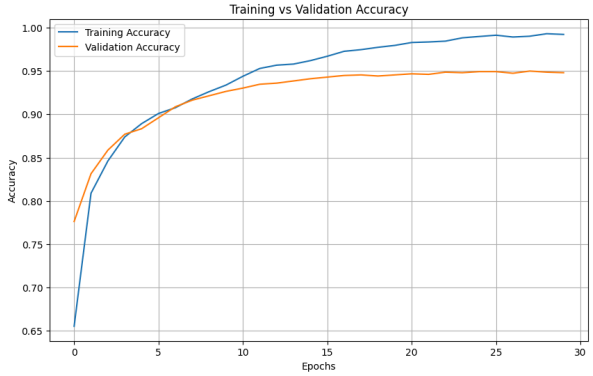
Model	**Acc**	**Pre**	**Recall**	**F1**	**Loss**	**Running time**	**Model Parameter (total and trainable)**
EfficientNetB0	0.9550	0.9639	0.9702	0.9670	0.1196	27.47 minutes	Total: 7,769,978 Trainable: 1,409 Non-trainable: 7,768,569
EfficientNetB2	0.9505	0.9510	0.9776	0.9642	0.1416	12.58 minutes	Total: 7,769,978 Trainable: 1,409 Non-trainable: 7,768,569
MobileNetV2	0.8781	0.9069	0.9145	0.9107	0.3710	3859.08 seconds	Total: 2,259,265 Trainable: 1,281 Non-trainable:2,257,984
ResetNet50	0.9068	0.9575	0.9031	0.9295	0.3404	25.14 minutes	Total Parameters: 23,589,761 ➢ Trainable Parameters: 2,049 ➢ Non-trainable Parameters:23,587,712

efficientNetB0

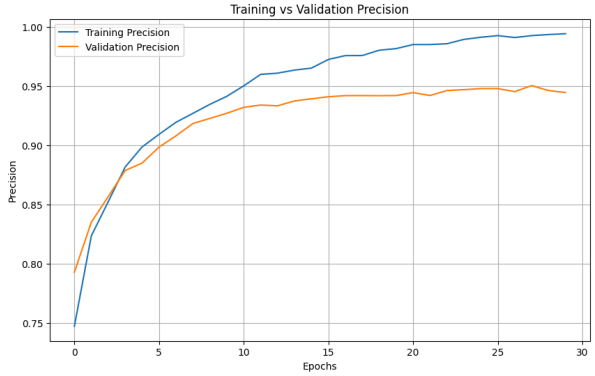
EfficientNetB2

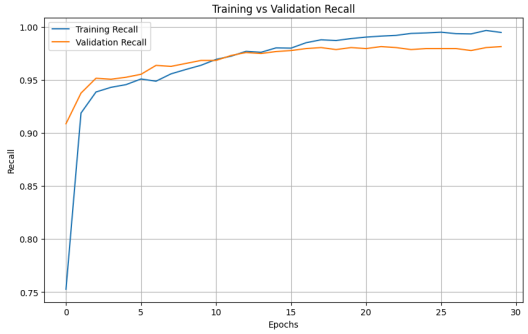
Finetuning_mobilenetV2

EfficientNetB2

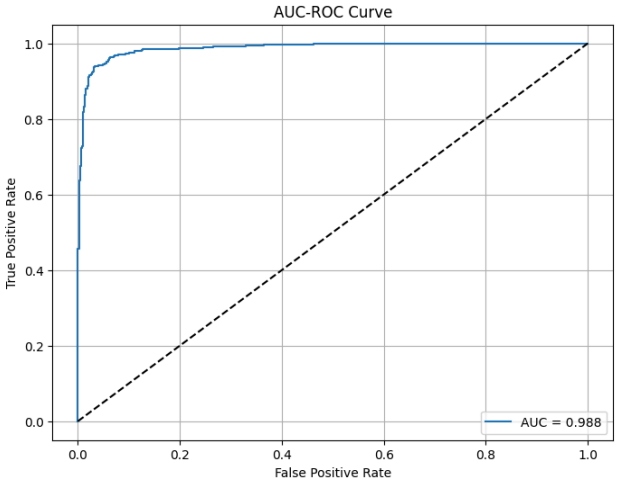


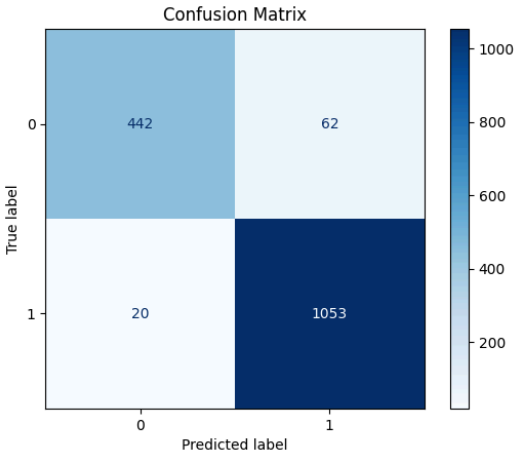




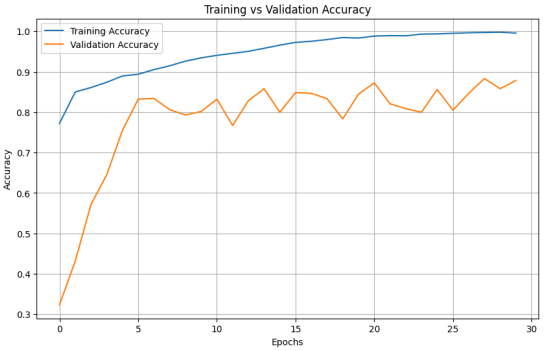




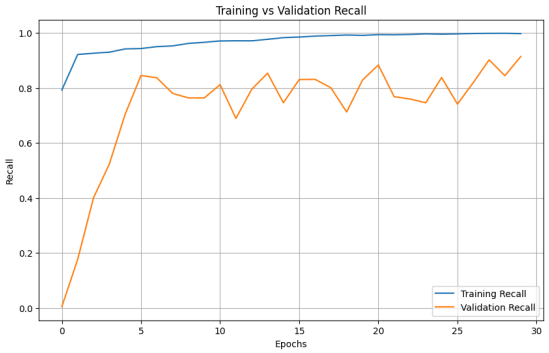


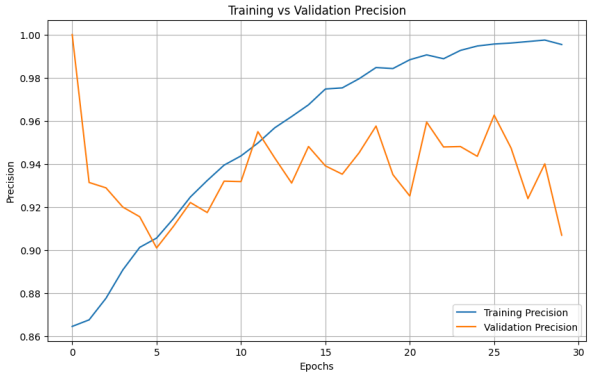


MobileNetV2:

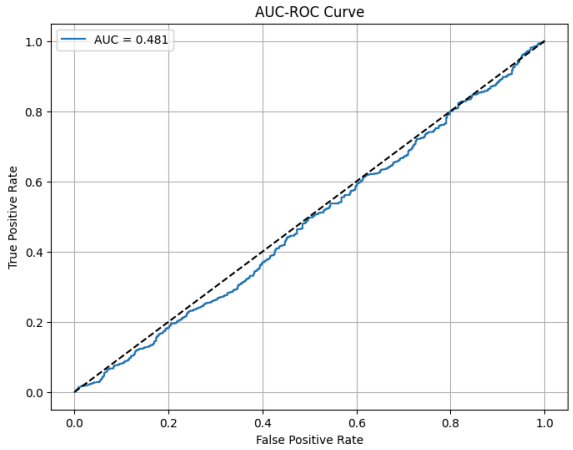




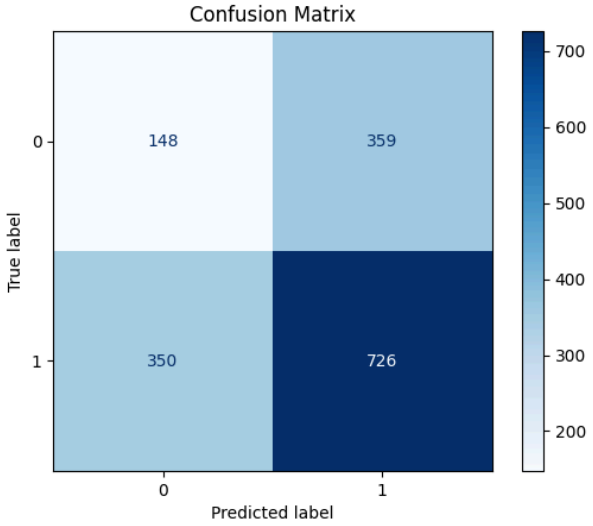

Training vs Validation Recall
Recall
Epochs
Training Recall
Validation Recall


Training vs Validation Precision
Precision
Epochs
Training Precision
Validation Precision

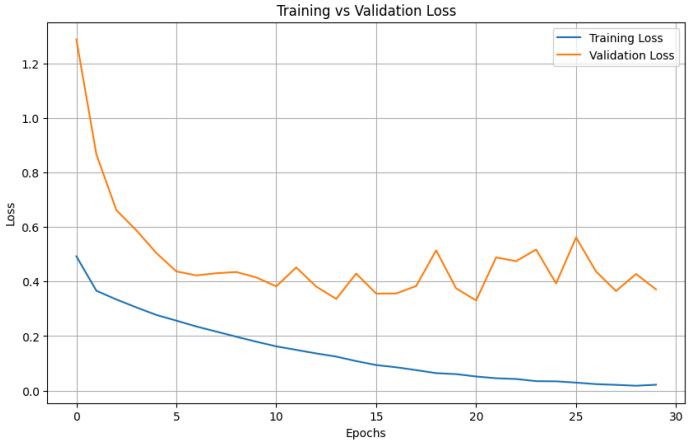

Training vs Validation F1 Score
F1 Score
Epochs
Training F1 Score
Validation F1 Score

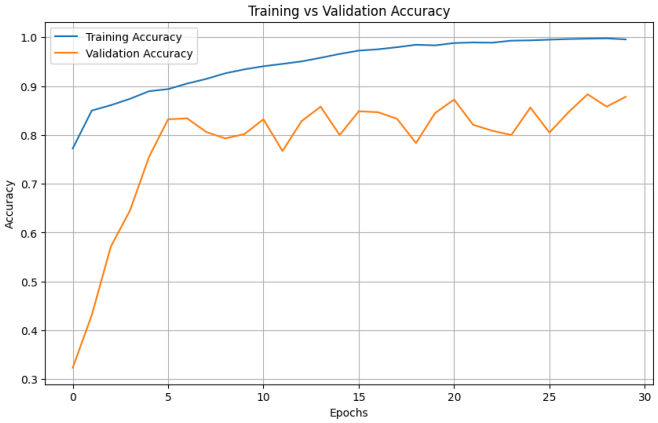

AUC-ROC Curve
AUC = 0.481
True Positive Rate
False Positive Rate

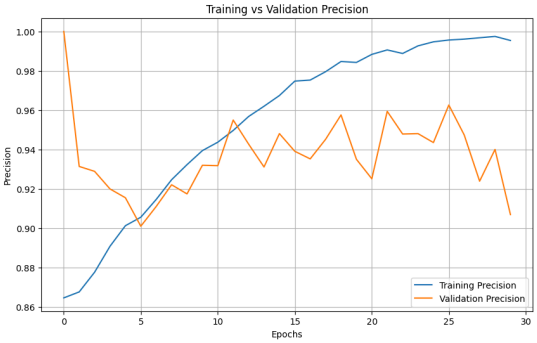
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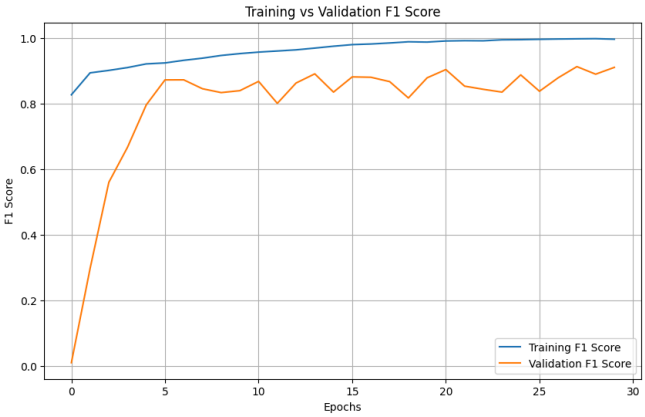

Confusion Matrix
148
359
350
726
True label
Predicted label

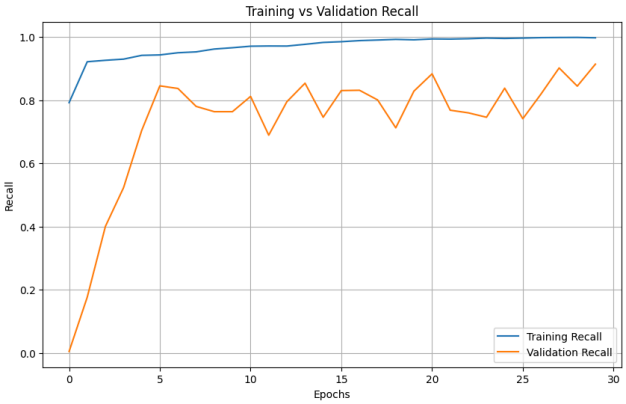
MobileNetV2:

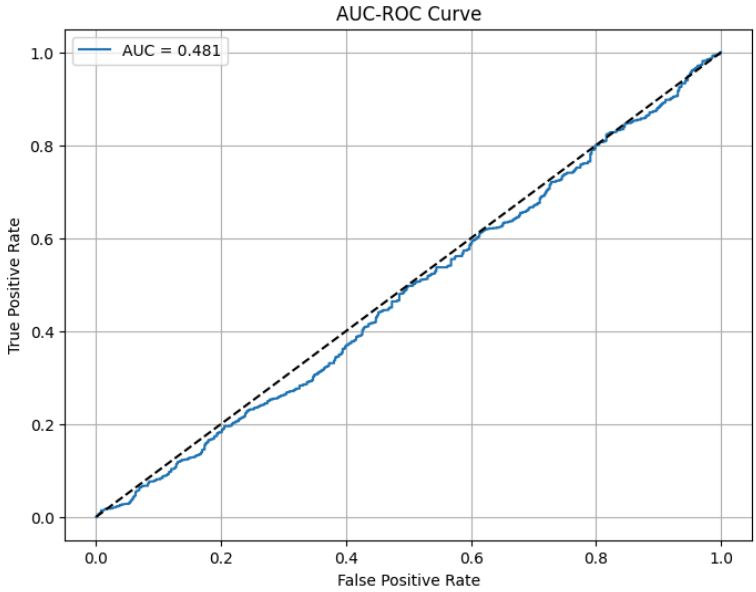


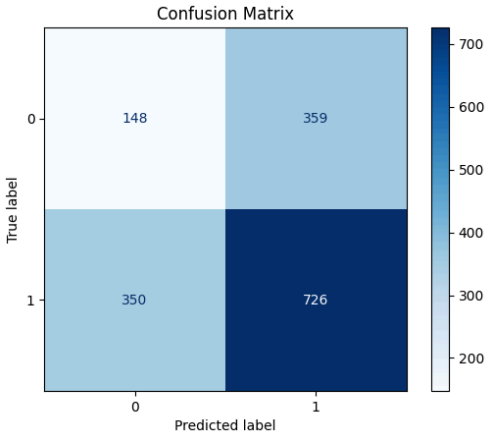





Training vs Validation F1 Score
F1 Score
Epochs
Training F1 Score
Validation F1 Score


Training vs Validation Recall
Recall
Epochs
Training Recall
Validation Recall


AUC-ROC Curve
AUC = 0.481
True Positive Rate
False Positive Rate



Hybrid/ensemble model:

Fine tuning
Unlock 20% layers

Epoch= 30 **Learning rate:0.0001** **No of Dropout Layer: 1** **Dropout value:0.3** **optimizer=Adam** **loss='binary_crossentropy'**							
Model	**Acc**	**Pre**	**Recall**	**F1**	**Loss**	**Running time**	**Model Parameter (total and trainable)**
EfficientNetB0 + MobileNetV2	0.9591	0.9647	0.9647	0.6831	0.1617	76 minutes 53.23 seconds	Total: 6,310,116 Trainable: 2,561 Non-trainable:6,307,555
EfficientNetB2 + MobileNetV2	0.9474	0.9626	0.9599	0.9613	0.1747	27.07 minutes	Total: 10,029,242 Trainable: 2,689 Non-trainable:10,026,553

EfficientNetB0+ RestNet50	0.9627	0.9730	0.9721	0.9726	0.1914	30 minutes 24.87 seconds	Total: 27,640,612 Trainable: 15,949,665 Non-trainable: 11,690,947

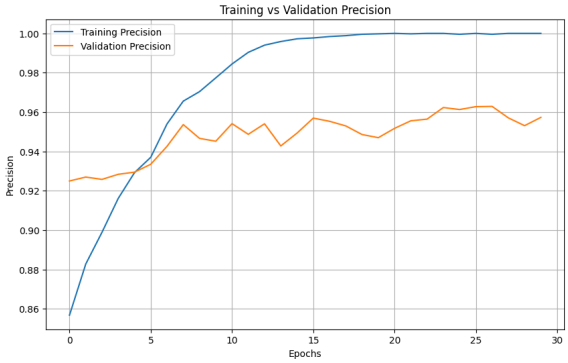
Without Fine tuning:

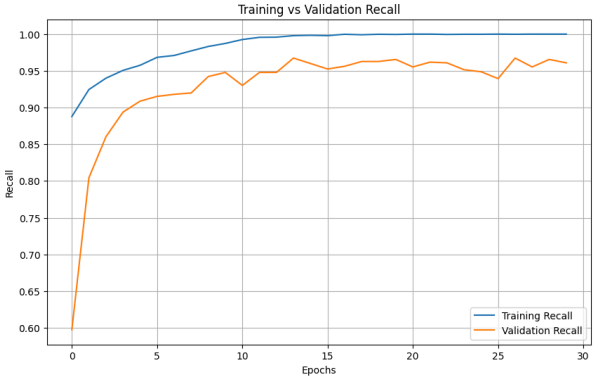
Epoch= 30 **Learning rate:0.0001** **No of Dropout Layer: 1** **Dropout value:0.3** **optimizer=Adam** **loss='binary_crossentropy'**							
Model	**Acc**	**Pre**	**Recall**	**F1**	**Loss**	**Running time**	**Model Parameter (total and trainable)**
EfficientNetB0 + MobileNetV2	0.9100	0.9066	0.9674	0.9360	0.2323	677.93 seconds	Total: 6,310,116 Trainable 2,561 Non-trainable:6,307,555
EfficientNetB2 + MobileNetV2	0.8222	0.8229	0.9414	0.8782	0.3835	819.04 seconds	Total: 10,029,242 Trainable: 2,689 Non-trainable: 10,026,553
EfficientNetB0+ RestNet50	0.6918	0.6894	0.9963	0.8149	0.9963	1057.50 seconds	Total: 27,640,612 Trainable: 3,329 Non-trainable:27,637,283

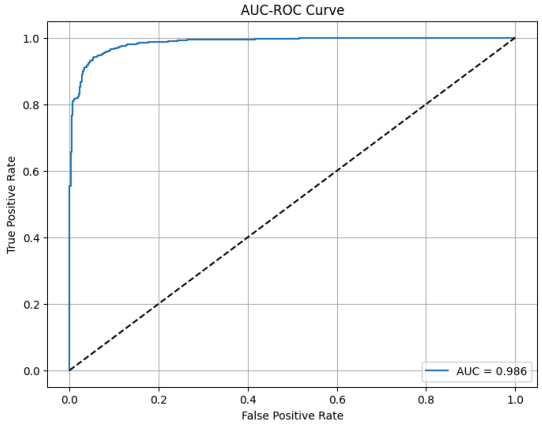
Ensemble model 1

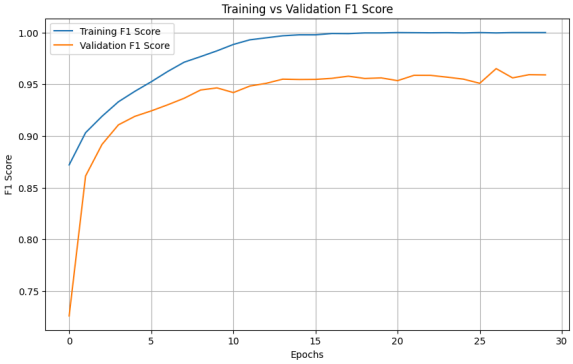

Training vs Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epochs

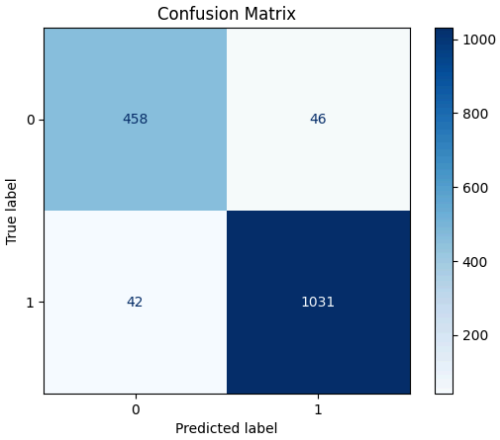

Training vs Validation Loss
Training Loss
Validation Loss
Loss
Epochs


Training vs Validation Precision
Training Precision
Validation Precision
Precision
Epochs


Training vs Validation Recall
Training Recall
Validation Recall
Recall
Epochs


AUC-ROC Curve
AUC = 0.986
True Positive Rate
False Positive Rate

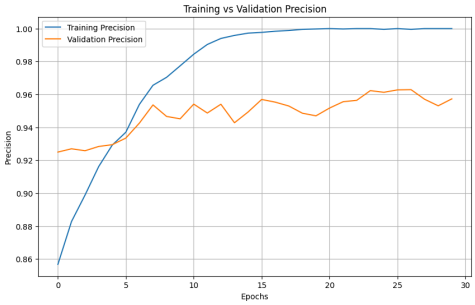

Training vs Validation F1 Score
Training F1 Score
Validation F1 Score
F1 Score
Epochs

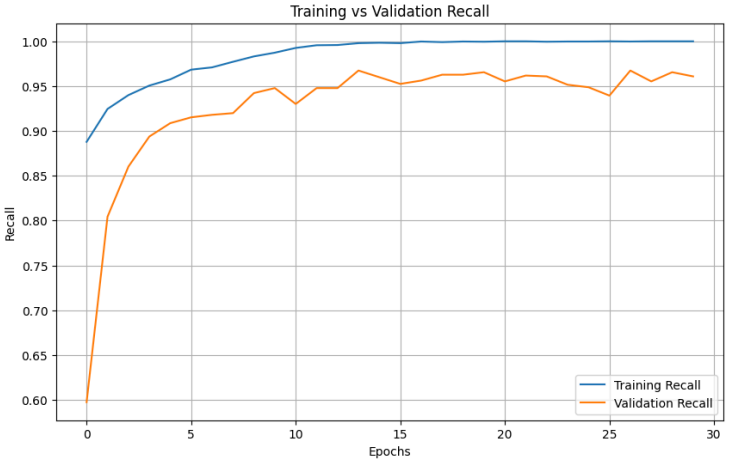


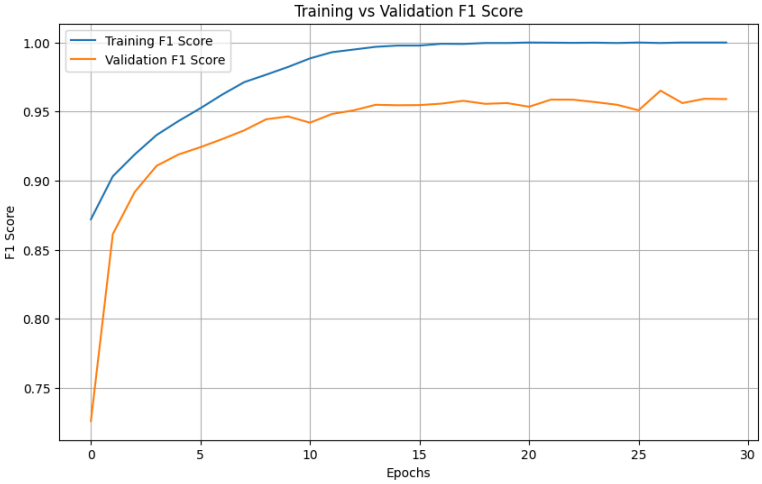
Ensemble Model 2

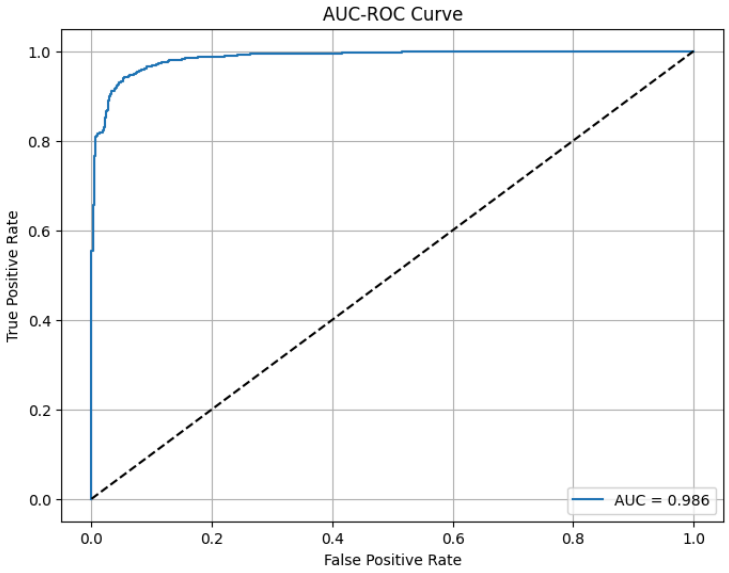


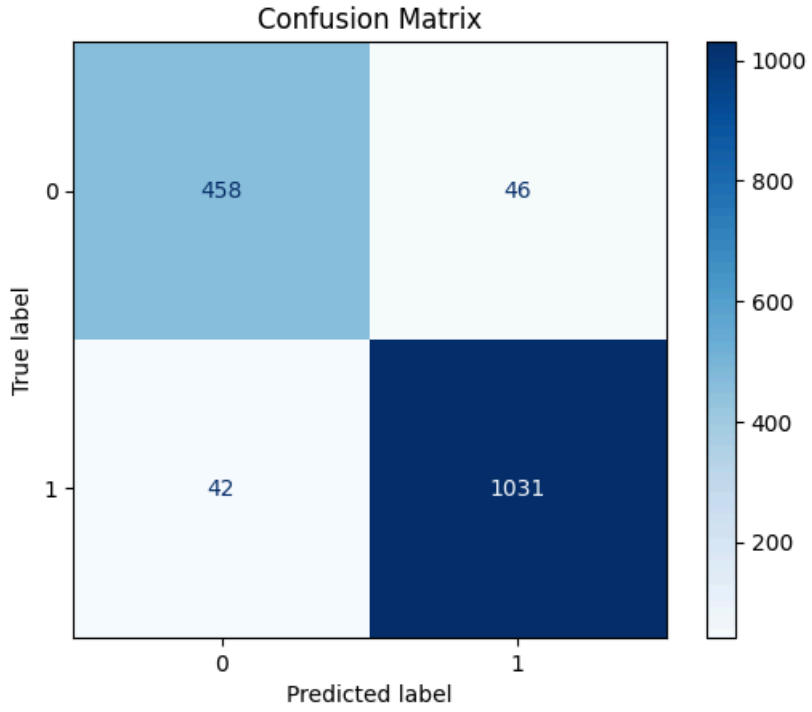




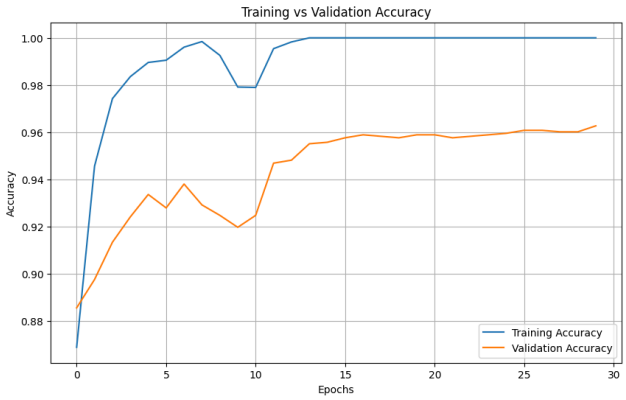

Training vs Validation Recall
Recall
Epochs
Training Recall
Validation Recall

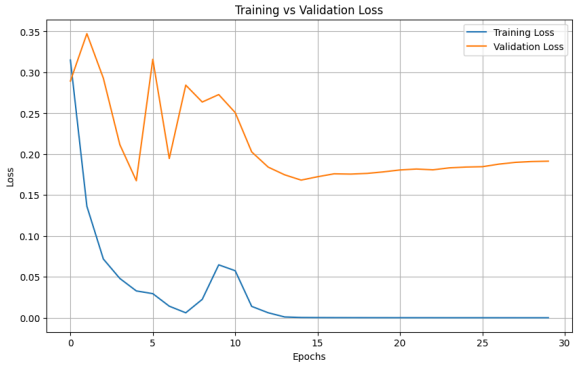

Training vs Validation F1 Score
F1 Score
Epochs
Training F1 Score
Validation F1 Score


AUC-ROC Curve
True Positive Rate
False Positive Rate
AUC = 0.986

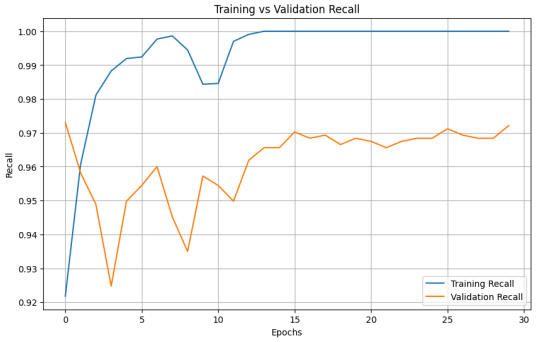


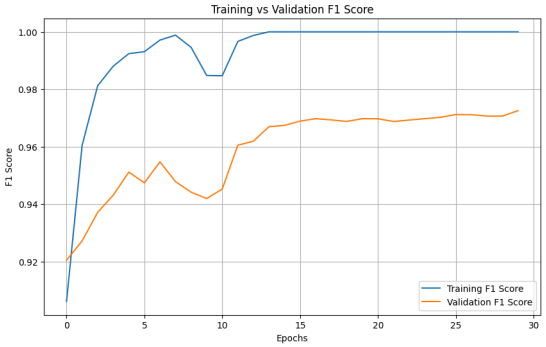
EfficientNetB0+RestNet50:

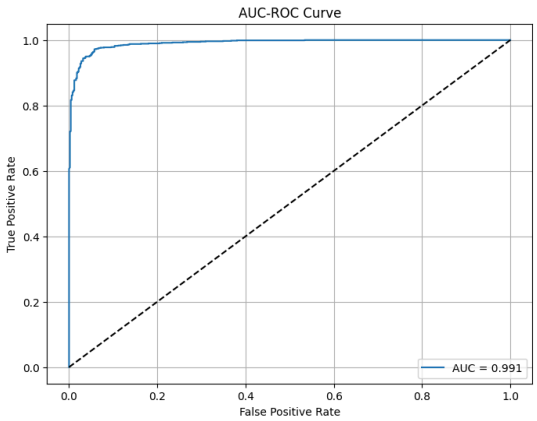


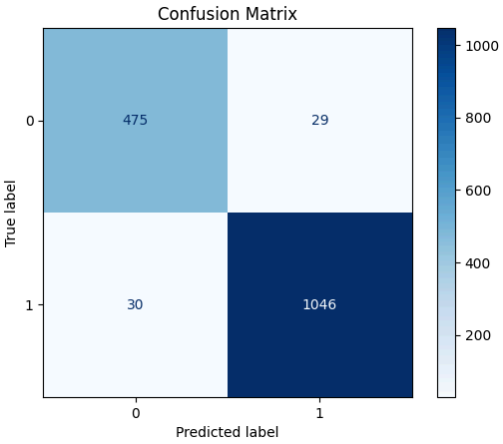

Training vs Validation Loss
Training Loss
Validation Loss
Loss
Epochs


Training vs Validation Precision
Training Precision
Validation Precision
Precision
Epochs


Training vs Validation Recall
Training Recall
Validation Recall
Recall
Epochs

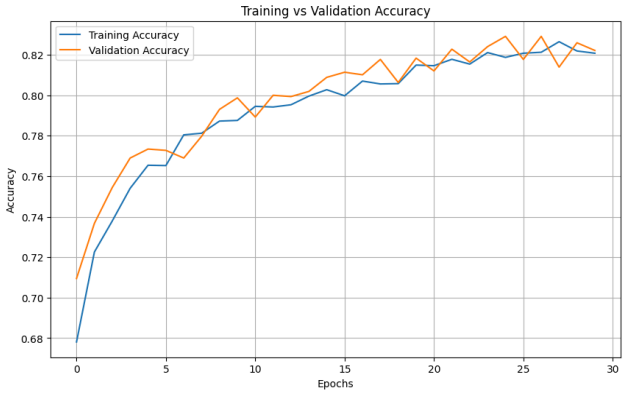

Training vs Validation F1 Score
Training F1 Score
Validation F1 Score
F1 Score
Epochs



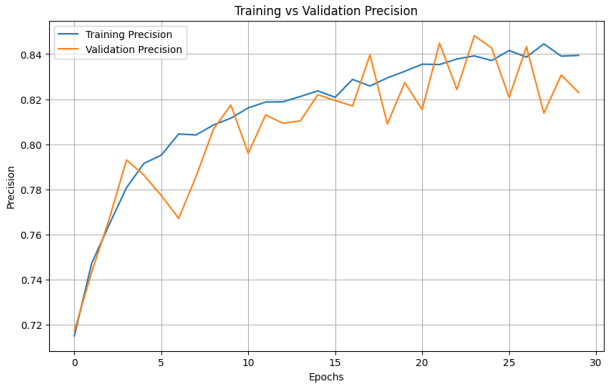


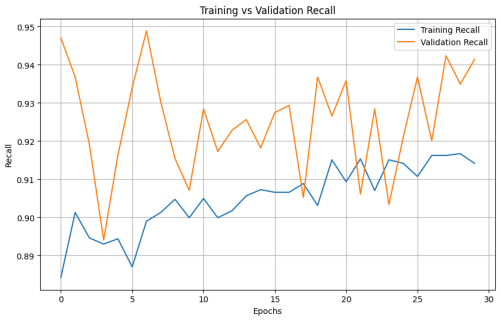
Without Fine-tuning:

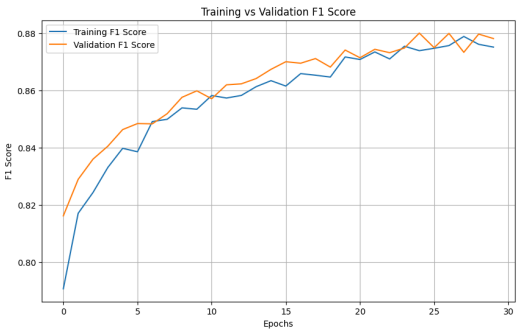
EfficientNetB2+MobileNetV2

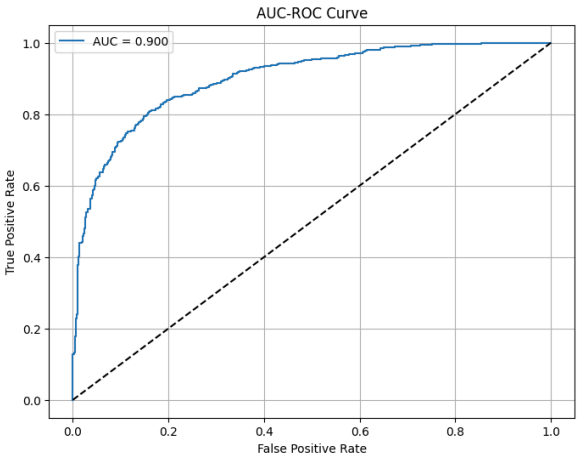

Training vs Validation Accuracy
Training Accuracy
Validation Accuracy
Accuracy
Epochs

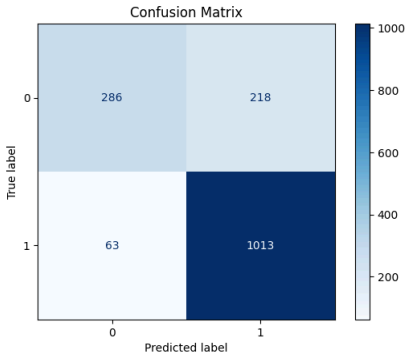

Training vs Validation Loss
Training Loss
Validation Loss
Loss
Epochs


Training vs Validation Precision
Training Precision
Validation Precision
Precision
Epochs

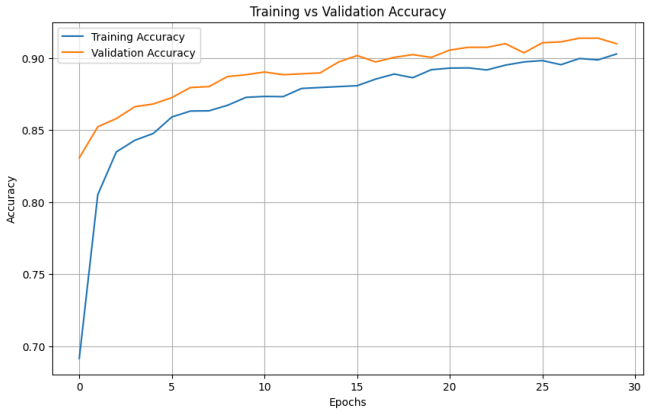

Training vs Validation Recall
Training Recall
Validation Recall
Recall
Epochs

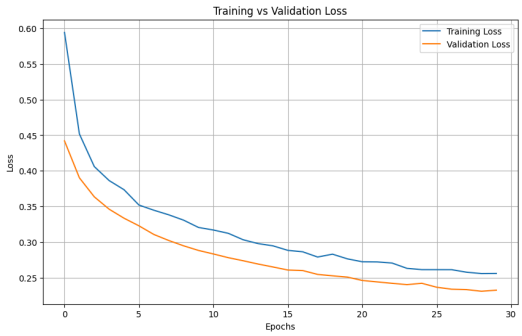

Training vs Validation F1 Score
Training F1 Score
Validation F1 Score
F1 Score
Epochs

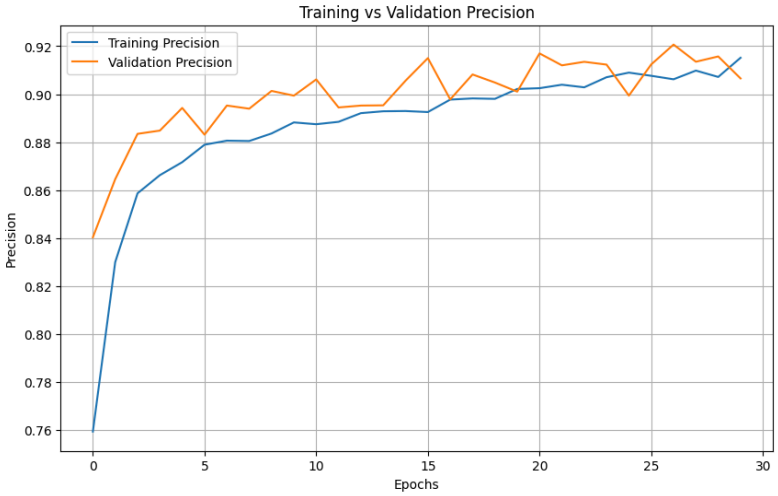

AUC-ROC Curve
AUC = 0.900
True Positive Rate
False Positive Rate

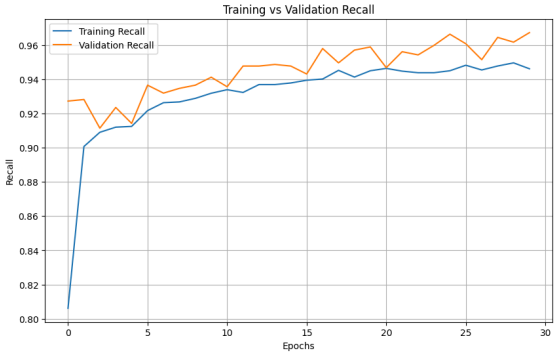


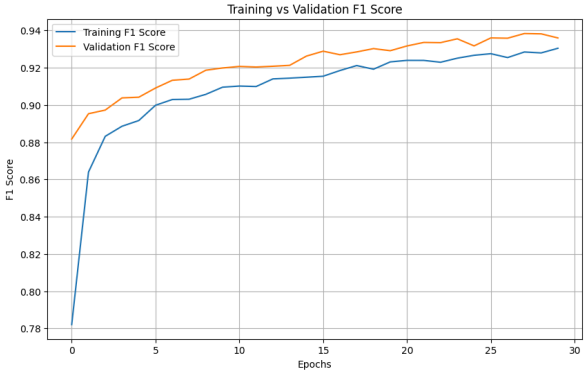
EfficientNetB0+MobileNetV2:

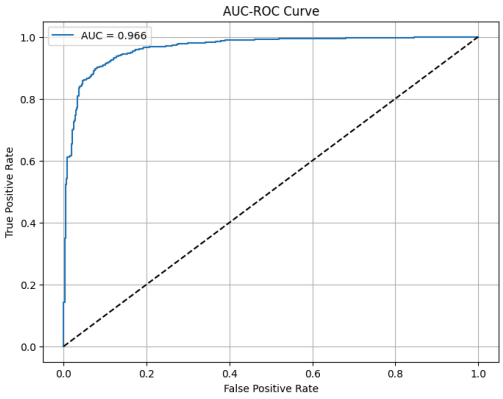


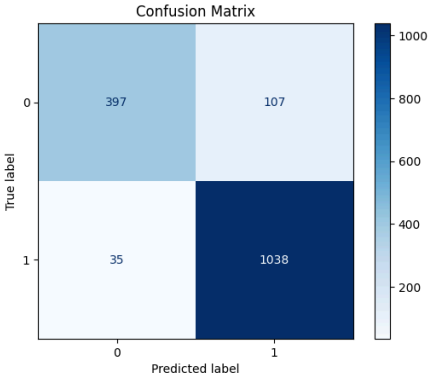



Training vs Validation Precision
Training Precision
Validation Precision
Precision
Epochs

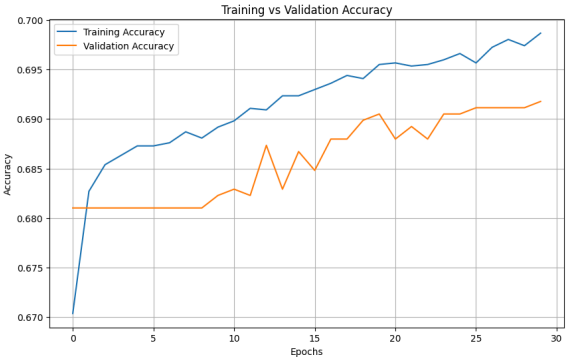

Training vs Validation Recall
Training Recall
Validation Recall
Recall
Epochs

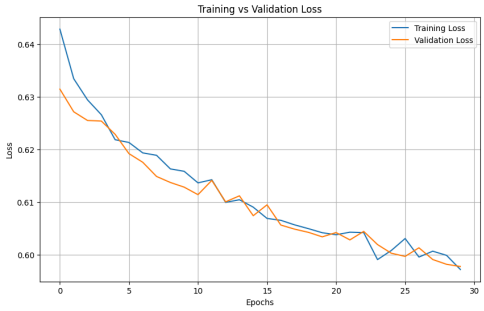

Training vs Validation F1 Score
Training F1 Score
Validation F1 Score
F1 Score
Epochs

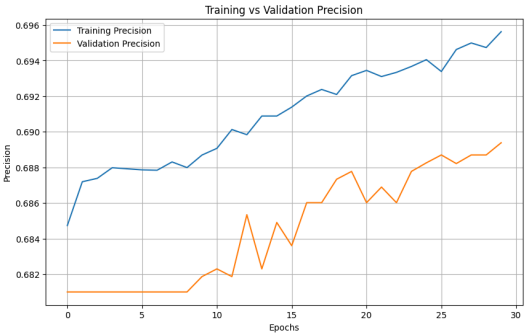

AUC-ROC Curve
AUC = 0.966
True Positive Rate
False Positive Rate

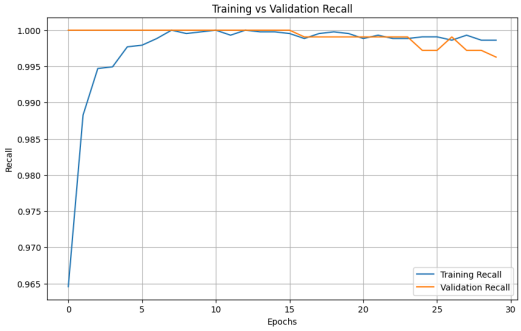


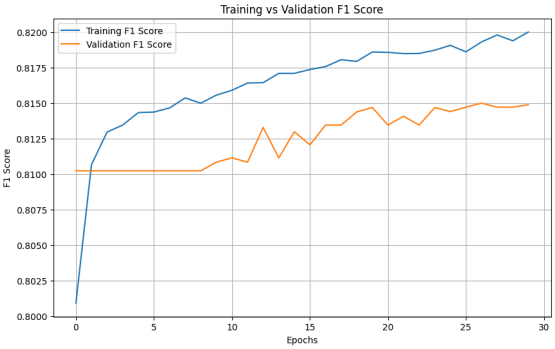
EfficientNetB0+RestNet50:

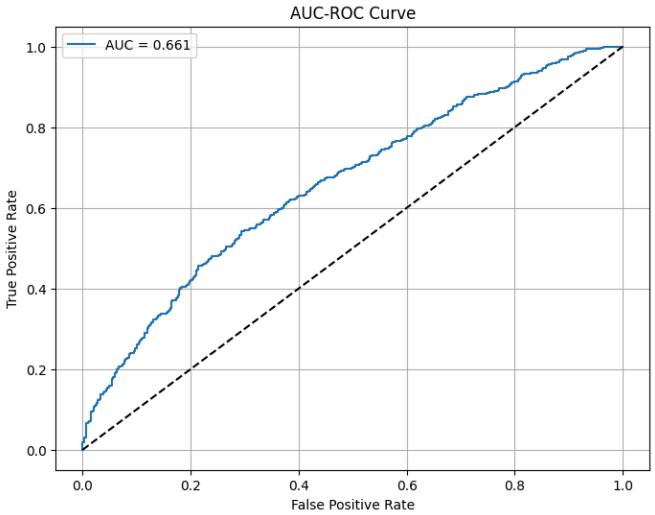


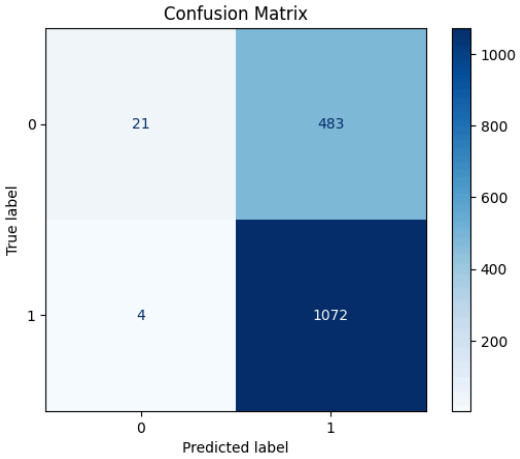



Training vs Validation Precision
Training Precision
Validation Precision
Precision
Epochs


Training vs Validation Recall
Training Recall
Validation Recall
Recall
Epochs


Training vs Validation F1 Score
Training F1 Score
Validation F1 Score
F1 Score
Epochs


AUC-ROC Curve
AUC = 0.661
True Positive Rate
False Positive Rate


Confusion Matrix
21
483
4
1072
0
1
True label
Predicted label
1000
800
600
400
200